

Iterative Workflows in LangGraph | Agentic AI using LangGraph

1. Recap

- **Sequential:** Steps execute one after another.
- **Parallel:** Independent steps execute at the same time.
- **Conditional:** The workflow chooses a path based on a condition.
- **Iterative:** The workflow **repeats steps until a condition is satisfied**.

Basic idea:

Generate → Evaluate → Improve → Evaluate → ... → Finish
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2. Plan of Action

The workflow is designed to repeatedly improve a result.

Typical plan:

1. Generate an initial result.
2. Evaluate the result.
3. Check whether it meets the required criteria.
4. If not, improve it.
5. Evaluate it again.
6. Stop when the result is acceptable.

The evaluation determines whether the workflow should **continue looping or finish**.

3. Building the Workflow

An iterative LangGraph workflow uses a **loop** between nodes.

Example:

Generate → Evaluate ↖ Improve ← If quality is insufficient ↓ Final Output ← If quality is sufficient

Important components:

- **State:** Stores the current result and other required information.
- **Nodes:** Perform generation, evaluation, and improvement.
- **Conditional edge:** Decides whether to continue or stop.
- **Loop:** Sends execution back to an earlier node.

This allows the workflow to progressively improve its output.

4. Testing

Testing verifies that the workflow behaves correctly during multiple iterations.

Check that:

- The workflow generates the expected result.
- Evaluation correctly determines quality.

- The workflow loops when improvement is needed.
- The workflow stops when the condition is satisfied.
- State is correctly updated between iterations.
- The workflow does not continue indefinitely.

Key takeaway: An iterative workflow uses **loops + conditions** to repeatedly perform and evaluate a task until the desired result is achieved.